

CARL-WSN: Context-Aware Reinforcement Learning for Hybrid Routing in Heterogeneous IoT-Driven Wireless Sensor Networks

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Abstract

Wireless Sensor Networks (WSNs) deployed in IoT-driven smart city environments carry heterogeneous application traffic with fundamentally different quality-of-service requirements. Existing routing protocols — including recent reinforcement learning (RL) based approaches — optimise a single objective such as energy efficiency or network lifetime without adapting routing behaviour to the urgency class of individual packets. This paper proposes CARL-WSN, a Context-Aware Reinforcement Learning protocol that uses a tabular Q-Learning agent to dynamically select the optimal routing paradigm — proactive, reactive, or hybrid — on a per-packet basis. The agent observes a three-dimensional context vector comprising application urgency class, normalised residual energy, and link stability score, and learns a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint hardcodes Class A emergency packets to proactive routing regardless of the agent's policy, guaranteeing sub-100 ms delivery at all times. CARL-WSN further integrates energy-aware relay selection and critical-node packet deferral to extend network lifetime without compromising latency guarantees. The Q-table requires only 216 bytes of RAM, enabling deployment on resource-constrained sensor hardware. CARL-WSN is evaluated in MATLAB against seven baseline

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protocols — including AODV, DSDV, ZRP, two domain-specific baselines, and two recent RL-based approaches (RLCR and FQ-UCR) — under four traffic scenarios across five independent seeds (160 runs). Results demonstrate that CARL-WSN achieves the lowest routing overhead of all evaluated protocols (2.27 control packets per data packet), an 82.5% reduction in Class A emergency latency compared to AODV, 55.5% lower overhead than RLCR, and 59.0% lower overhead than FQ-UCR, while achieving the best or near-best performance on six of eight evaluated metrics including network lifetime (FND = 749), energy efficiency (0.053 J/round), throughput (85.7 pkts/round), and energy balance (Gini = 0.185). The learned Q-table converges within 500 rounds and the agent autonomously discovers context-dependent routing policies that adapt overhead to traffic class composition — a capability absent in all seven baselines.

Keywords: Wireless sensor networks, Reinforcement learning, Q-Learning, Context-aware routing, Hybrid routing protocol, IoT, Smart city, Energy-aware relay selection, Heterogeneous traffic

1. Introduction

The proliferation of Internet of Things (IoT) devices in smart city infrastructure has transformed Wireless Sensor Networks (WSNs) from single-application monitoring systems into multi-service platforms that simultaneously carry traffic with fundamentally different quality-of-service (QoS) requirements [1, 2]. A single WSN deployment may generate emergency alerts requiring sub-100 ms delivery, periodic telemetry tolerating delays of tens of seconds, and event-driven uploads with best-effort requirements. This traffic heterogeneity demands routing protocols that adapt their behaviour to the urgency class of individual packets — a capability absent from both classical and recent learning-based routing approaches.

Routing protocols for WSNs fall into three paradigms, each with inherent trade-offs. Proactive protocols such as DSDV [3] maintain routing tables continuously, delivering low latency at the cost of high energy overhead from periodic broadcasts. Reactive protocols such as AODV [4] and DSR [5] discover routes on demand, conserving energy in low-traffic conditions but introducing route-discovery delays of 20–100 ms that violate emergency latency constraints [6]. Hybrid protocols such as ZRP [7] partition the network into zones and apply different modes within and across boundaries, but use a fixed

zone radius that does not adapt to node energy state or traffic urgency.

Recent work has applied reinforcement learning (RL) to WSN routing with promising results. Daanoun and Baghdad [8] proposed RLCR, a Q-Learning-based clustering and routing protocol that optimises cluster head selection and inter-cluster forwarding for energy efficiency. Wang and Duan [9] proposed FQ-UCR, combining fuzzy logic for unequal cluster formation with Q-Learning for inter-cluster routing. While both protocols demonstrate the potential of RL for adaptive WSN routing, they share a fundamental limitation: routing decisions are made based on network-level metrics (residual energy, distance to base station) without any awareness of the application-level urgency class of the packet being routed. An emergency alert and a routine temperature reading receive identical routing treatment — a systematic inefficiency that leads to either wasted energy on low-urgency traffic or dangerous delays on safety-critical packets.

A 2025 comprehensive survey by Khedr et al. [10] explicitly identified dynamic application-context-based hybrid mode switching as an open research direction in IoT-driven WSNs, noting that reactive and hybrid context-aware approaches have received significantly less attention than proactive ones. No published protocol uses reinforcement learning to select between fundamentally different routing paradigms — proactive, reactive, and hybrid — on a per-packet basis according to application urgency. This paper addresses that gap directly.

The main contributions of this paper are as follows:

- A formal Markov Decision Process (MDP) formulation for routing paradigm selection in heterogeneous IoT-WSNs, with an $18\text{-state} \times 3\text{-action}$ state-action space derived from a three-dimensional context vector comprising application urgency class U , normalised residual energy E_r , and link stability score L_s .
- CARL-WSN, a Q-Learning-based routing protocol that learns the optimal mapping from context states to routing paradigms (proactive, reactive, or hybrid) through online interaction with the network environment, using a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint hardcodes Class A emergency packets to proactive routing, guaranteeing sub-100 ms delivery regardless of the agent’s learned policy.

- An energy-aware relay selection mechanism combined with critical-node packet deferral that extends network lifetime by preferring high-residual-energy next-hop nodes and deferring non-urgent packets when the transmitting node is near energy depletion — a context-aware lifetime extension capability that only urgency-aware protocols can implement.
- Comprehensive evaluation against seven baselines — three classical protocols (AODV, DSDV, ZRP), two domain-specific baselines (EH-Routing, MSLBA), and two recent RL-based protocols (RLCR, FQ-UCR) — across four traffic scenarios and five independent seeds (160 runs), demonstrating that CARL-WSN achieves the lowest routing overhead of all evaluated protocols while maintaining competitive or superior performance on six of eight metrics.

The remainder of this paper is organised as follows. Section 2 reviews related work on hybrid, context-aware, and RL-based routing in WSNs. Section 3 presents the system model. Section 4 describes the CARL-WSN protocol design, including the MDP formulation and Q-Learning mechanism. Section 5 presents the simulation setup. Section 6 presents and analyses the results. Section 7 discusses key findings, trade-offs, and limitations. Section 8 concludes the paper.

2. Related Work

2.1. Proactive and Reactive Routing Protocols

Routing protocols for WSNs have been extensively surveyed and classified [11]. Proactive protocols maintain routing tables continuously through periodic broadcasts. DSDV [3] established the foundational destination-sequenced distance-vector approach, guaranteeing loop-free routes at all times. OLSR [12] extended this paradigm through multipoint relay selection, substantially reducing flooding overhead in dense networks. While proactive protocols deliver consistently low latency by eliminating route discovery delays, their continuous maintenance overhead leads to significant energy waste on low-traffic nodes [11].

Reactive protocols discover routes exclusively on demand. AODV [4] pioneered this approach by initiating route requests only when data must be transmitted, conserving energy in sparse or low-traffic conditions. However, the route discovery delay of 20–100 ms introduced by RREQ/RREP

flooding renders reactive protocols unsuitable for latency-sensitive applications [6]. A comparative analysis by Nurul Hidayah et al. [13] confirmed that neither proactive nor reactive protocols alone achieve optimal performance across varying network conditions, motivating the need for adaptive hybrid approaches. Recent work [14] demonstrated that AODV, DSDV, and DSR all exhibit significant performance degradation under heterogeneous traffic loads, highlighting the fundamental limitation of single-mode routing for IoT-driven smart city deployments.

2.2. Hybrid Routing Protocols

Hybrid protocols attempt to combine the latency advantages of proactive routing with the energy efficiency of reactive routing. ZRP [7] pioneered zone-based hybrid routing by maintaining proactive routes within a fixed-radius zone and using reactive inter-zone routing for destinations beyond the zone boundary. While ZRP reduces control overhead compared to purely proactive protocols, its fixed zone radius is configured statically and does not adapt to node energy state or traffic urgency.

Subsequent hybrid approaches incorporated additional optimisation dimensions. Chaurasiya et al. [15] proposed a hybrid clustering technique for multilevel heterogeneous IoT WSNs, combining cluster-based organisation with multi-hop routing to reduce energy consumption. Lei [16] proposed a hybrid energy-aware approach combining particle swarm optimisation with fuzzy clustering for IoT routing, demonstrating 52% energy reduction compared to DEEC, though at the cost of requiring offline training data. Nadanam et al. [17] recently proposed a quantum-optimised hybrid routing framework achieving 30% energy reduction, though the computational complexity of quantum optimisation makes it unsuitable for resource-constrained sensor hardware.

A common limitation across all these hybrid approaches is that mode selection or routing decisions are made based on a single optimisation objective — typically energy or load — without awareness of the application-level urgency of the traffic being routed.

2.3. Context-Aware Routing in IoT-WSNs

Context-aware computing for IoT was formally defined by Perera et al. [18], who identified five contextual dimensions — identity, location, activity, time, and relations — as fundamental to intelligent IoT decision making. This foundational work motivates the use of contextual information in routing

126 decisions, though the specific application to RL-based WSN hybrid routing
127 mode selection had not been addressed until the present work.

128 Context-aware routing has since been applied across several IoT and WSN
129 domains. Monteiro et al. [19] proposed context-aware network selection in
130 heterogeneous wireless networks, demonstrating that context-driven decisions
131 outperform static policies, but their work focused on network selection rather
132 than intra-WSN routing mode switching. Sithik and Kumar [20] proposed
133 a multi-context aware parent selection scheme for the RPL protocol using
134 grey set theory, incorporating multiple contextual parameters into routing
135 metric computation. However, their approach targets RPL parent selection
136 exclusively and does not implement dynamic switching between fundamentally
137 different routing paradigms based on application urgency class. Karagiannis
138 et al. [21] proposed context-aware routing for fog computing systems, reducing
139 latency by 23% and hop count by 73%, though their work operates at the fog
140 layer rather than the WSN sensor layer and does not model heterogeneous
141 traffic classes with conflicting QoS requirements. Hasan et al. [22] proposed
142 a context-aware trust prediction framework for opportunistic IoT routing
143 using Bayesian inference, achieving 88.9% delivery ratio, though the trust-
144 based context dimension differs fundamentally from the urgency-energy-
145 stability context vector proposed in CARL-WSN. Akpakwu et al. [23] proposed
146 a context-aware congestion control scheme for CoAP/UDP IoT networks,
147 applying context sensing to transport-layer decisions rather than routing.

148 Smart city IoT routing has also received significant attention. Karunk-
149 uzhali et al. [24, 25] proposed QoS-aware routing frameworks for IoT-WSNs
150 in smart cities, demonstrating that multi-class service differentiation improves
151 network performance. However, none of these works implement dynamic rout-
152 ing paradigm switching based on per-packet application urgency. El-Fouly et
153 al. [26] proposed environment-aware real-time routing for multi-sink WSNs
154 in smart cities, incorporating deadline constraints and link quality metrics.
155 While this work shares the multi-sink topology with our prior work [27]
156 and considers real-time delivery requirements. Smart city IoT routing has
157 also been addressed by Hanif et al. [28] through opportunistic IoT-WSN
158 integration, Inga et al. [29] through WSN sizing and routing optimisation,
159 and Ekler et al. [30] through energy-aware IoT routing algorithms. However,
160 none of these works implement dynamic routing paradigm switching based on
161 per-packet application urgency. it does not model heterogeneous application
162 urgency classes or implement context-driven mode switching.

2.4. Reinforcement Learning for WSN Routing

The application of reinforcement learning to WSN routing has emerged as one of the most active research frontiers in recent years [31, 32, 33]. RL-based protocols offer the ability to learn routing policies through online interaction with the network environment, adapting to changing conditions without requiring explicit models or predefined rules.

Daanoun and Baghdad [8] proposed RLCR, a reinforcement learning-based clustering and routing protocol for heterogeneous WSNs. RLCR uses Q-Learning in two phases: first for cluster head (CH) selection based on residual energy, distance to the base station, and node density; and second for inter-cluster multi-hop routing where each CH learns the optimal next-hop CH toward the base station. RLCR demonstrated significant lifetime improvements over LEACH and DEEC variants. However, RLCR does not differentiate traffic by application urgency class — all packets receive identical routing treatment regardless of whether they carry emergency alerts or routine telemetry. Furthermore, RLCR’s RL agent operates exclusively within the clustering paradigm and does not consider switching between fundamentally different routing modes.

Wang and Duan [9] proposed FQ-UCR, a hybrid approach combining fuzzy logic (FL) for unequal cluster head selection with Q-Learning for inter-cluster routing. FQ-UCR addresses the hotspot problem by forming smaller clusters near the base station and larger clusters at greater distances, using a fuzzy inference system to compute CH selection probability. The Q-Learning agent then selects the optimal forwarding CH for inter-cluster data relay. FQ-UCR demonstrated improved energy efficiency and lifetime compared to EEUC and CHEF. Similar to RLCR, however, FQ-UCR does not model heterogeneous traffic classes or adapt routing behaviour to application urgency.

Sudha et al. [34] combined Q-Learning with machine learning for cluster head selection and routing in 5G-enabled WSNs, demonstrating improved throughput over LEACH, but without considering per-packet application urgency as a routing criterion. Saadallah and Alabady [35] identified energy-efficient context-sensitive routing as an open challenge in their comprehensive IoT routing study, while Kumar et al. [36] confirmed through bibliometric analysis that application-context awareness remains an emerging research gap in energy-efficient IoT-WSN routing.

Table 1 summarises the key distinguishing features of CARL-WSN compared to the most closely related works. CARL-WSN is the only protocol that

Table 1: Comparison of CARL-WSN with Related Routing Works

Protocol	AI/ML method	Context dims	Mode switching	Traffic classes	Urgency aware	Energy-aware relay	Paradigm switching
RLCR [8]	Q-Learning	3	CH selection	1	No	No	No
FQ-UCR [9]	Fuzzy + Q-Learning	3	CH selection	1	No	No	No
Sithik & Kumar [20]	Grey set theory	2	RPL parent	1	No	No	No
Karagiannis et al. [21]	None	2	Fog forwarding	1	No	No	No
Hasan et al. [22]	Bayesian	1 (trust)	Route selection	1	No	No	No
ZRP [7]	None	1 (zone)	Hybrid fixed	1	No	No	No
CARL-WSN (Ours)	Q-Learning	3	Per-packet RL	3	Yes	Yes	Yes

combines RL-based learning with per-packet application urgency awareness for full routing paradigm switching.

2.5. Research Gaps

The preceding review reveals four specific gaps that motivate CARL-WSN. First, Khedr et al. [10] conducted the first comprehensive survey of context-aware routing in IoT-driven WSNs and explicitly identified dynamic application-context-based mode switching as an open research direction, noting that reactive and hybrid context-aware methods have received significantly less attention than proactive approaches. Second, existing RL-based WSN routing protocols (RLCR [8], FQ-UCR [9]) apply Q-Learning within a single routing paradigm (clustering) without awareness of application urgency class — the RL agent cannot distinguish between emergency and routine traffic. Third, Chu et al. [37], Khanna et al. [38], and Prashanth et al. [39] all noted that routing adaptation in heterogeneous WSNs remains dominated by single-objective optimisation approaches that cannot simultaneously satisfy the conflicting requirements of heterogeneous application traffic classes. Fourth, no published protocol uses RL to learn which routing paradigm (proactive, reactive, or hybrid) to use for each packet based on a joint context vector of urgency, energy, and link stability. Fifth, no RL-based WSN routing protocol incorporates a safety constraint that guarantees latency bounds for critical traffic regardless of the learned policy — a requirement for safety-critical IoT deployments.

3. System Model

3.1. Network Topology

Consider a WSN comprising N heterogeneous sensor nodes deployed randomly over a square area of side length A m according to a uniform distribution. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of all nodes. A base

station (BS) is located at the centre of the deployment area, and a secondary sink node is positioned at the right boundary, consistent with the multi-sink topology in our prior work [27]. Both the BS and sink are assumed to have unconstrained energy. Each sensor node $i \in \mathcal{N}$ is characterised by a fixed position (p_i^x, p_i^y) and a heterogeneous initial energy E_0^i drawn from a uniform distribution $E_0^i \sim \mathcal{U}(E_{\min}, E_{\max})$, reflecting the reality of diverse sensor hardware in smart city deployments [40].

A node i is considered alive if its residual energy $E_i(t) > 0$ at simulation round t , and dead otherwise. Dead nodes neither transmit nor receive. The network is considered operational as long as at least 10% of nodes remain alive.

3.2. Traffic Model

Smart city IoT deployments generate heterogeneous application traffic with fundamentally different QoS requirements. This paper models three traffic classes simultaneously:

- **Class A — Emergency traffic:** Generated by safety-critical sensors such as fire detectors, intrusion alarms, and accident monitors. Modelled as a Poisson process with arrival rate λ_A . Strict latency requirement: end-to-end delay must not exceed 100 ms.
- **Class B — Periodic telemetry:** Generated by environmental sensors reporting temperature, humidity, air quality, and energy consumption. Modelled as a deterministic periodic process with one packet per minute. Tolerates delays up to 30 seconds.
- **Class C — Event-driven uploads:** Generated by motion detectors, occupancy sensors, and camera triggers. Modelled as a Poisson process with arrival rate λ_C . Best-effort delivery with no hard latency constraint.

Each packet carries a class tag $U \in \{A, B, C\}$ in its header, which is read by the CARL-WSN context classifier at the transmitting node before routing decisions are made.

3.3. Channel and Energy Model

Wireless communication is modelled using the first-order radio energy model [41], which captures the dominant sources of energy consumption in

258 sensor nodes: electronics and amplification. The energy consumed by node i
 259 to transmit an L -bit packet to a receiver at distance d is:

$$E_{tx}(L, d) = \begin{cases} L \cdot E_{elec} + L \cdot \varepsilon_{fs} \cdot d^2 & d < d_0 \\ L \cdot E_{elec} + L \cdot \varepsilon_{mp} \cdot d^4 & d \geq d_0 \end{cases} \quad (1)$$

260 where $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} = 87.71$ m is the crossover distance between free-
 261 space and multipath propagation models. The energy consumed to receive
 262 an L -bit packet is:

$$E_{rx}(L) = L \cdot E_{elec} \quad (2)$$

263 The residual energy of node i after a transmission round is:

$$E_i(t+1) = E_i(t) - E_{tx}(L, d) - E_{rx}(L) \quad (3)$$

264 subject to $E_i(t) \geq 0$.

265 3.4. Link Stability Model

266 To capture dynamic link quality variations, CARL-WSN maintains a per-
 267 node link stability score $L_s^i(t) \in [0, 1]$, computed as an Exponential Weighted
 268 Moving Average (EWMA) of the ACK success history over a sliding window
 269 of W most recent transmissions:

$$L_s^i(t) = \sum_{k=0}^{W-1} w_k \cdot a_k^i \quad (4)$$

270 where $a_k^i \in \{0, 1\}$ is the ACK outcome of the k -th most recent transmission
 271 from node i , and $w_k = \alpha(1 - \alpha)^k/Z$ are normalised exponentially decaying
 272 weights with smoothing factor $\alpha = 0.3$ and normalisation constant $Z =$
 273 $\sum_{k=0}^{W-1} (1 - \alpha)^k$.

274 3.5. Energy-Aware Relay Selection Model

275 CARL-WSN introduces an energy-aware relay selection mechanism that
 276 extends network lifetime by preferring next-hop nodes with higher residual
 277 energy. When forwarding a packet, node i selects the next-hop relay j^* from
 278 its set of alive neighbours $\mathcal{N}_i = \{j : d_{ij} \leq R_{tx}, E_j(t) > 0\}$ by maximising a
 279 composite score:

$$j^* = \arg \max_{j \in \mathcal{N}_i} \left[w_p \cdot \frac{d_{iBS} - d_{jBS}}{d_{iBS}} + w_e \cdot \frac{E_j(t)}{E_0^j} \right] \quad (5)$$

where d_{iBS} and d_{jBS} are the distances from nodes i and j to the base station respectively, $w_p = 0.4$ is the progress weight, and $w_e = 0.6$ is the energy weight. The first term encourages geographic progress toward the BS, while the second term favours relays with higher residual energy ratios, distributing the forwarding load away from energy-depleted nodes. Only candidates with positive progress ($d_{jBS} < d_{iBS}$) are considered, ensuring packets always advance toward the BS.

3.6. Critical-Node Packet Deferral

When a node's residual energy ratio falls below a critical threshold $E_r^i < T_{defer} = 0.15$, CARL-WSN defers transmission of non-urgent packets (Class B and Class C) to preserve the node's remaining energy for potential emergency traffic. Deferred packets incur only a minimal listening energy cost of $0.1 \cdot L \cdot E_{elec}$. Class A emergency packets are never deferred regardless of node energy state — the safety constraint takes absolute priority. This deferral mechanism is a unique capability of urgency-aware protocols: baseline protocols that do not distinguish traffic classes cannot selectively defer low-priority traffic to extend node lifetime.

3.7. Performance Metrics

Eight metrics are used to evaluate protocol performance:

Network lifetime: The First Node Death round $r_{FND} = \min\{t : \exists i, E_i(t) = 0\}$ and the Half Node Death round $r_{HND} = \min\{t : |\{i : E_i(t) = 0\}| \geq N/2\}$.

Packet Delivery Ratio:

$$PDR = \frac{D_{delivered}}{D_{total}} \quad (6)$$

Energy balance (Gini coefficient):

$$G = \frac{2 \sum_{i=1}^N i \cdot c_i}{N \sum_{i=1}^N c_i} - \frac{N+1}{N} \quad (7)$$

where c_i is the energy consumed by node i , sorted in ascending order ($c_1 \leq c_2 \leq \dots \leq c_N$). A lower G indicates more uniform energy consumption.

306 **Routing overhead:**

$$\Omega = \frac{C_{ctrl}}{D_{total}} \quad (8)$$

307 where C_{ctrl} is the total number of routing control packets generated during
308 the simulation.

309 **Additional metrics:** Mean Class A latency, average energy consumed
310 per round, and network throughput (packets delivered per round).

311 4. CARL-WSN Protocol Design

312 4.1. Design Motivation

313 Existing RL-based WSN routing protocols apply Q-Learning within a
314 single routing paradigm. RLCR [8] uses Q-Learning for cluster head selection
315 and inter-cluster forwarding — both decisions operate within the clustering
316 paradigm. FQ-UCR [9] similarly applies Q-Learning only to select the next
317 forwarding cluster head. In both protocols, the fundamental routing mecha-
318 nism (clustering with multi-hop inter-cluster relay) never changes regardless
319 of traffic urgency.

320 CARL-WSN addresses a categorically different problem: *which routing*
321 *paradigm should be used for each packet?* The RL agent learns to select
322 between proactive, reactive, and hybrid routing — three fundamentally differ-
323 ent mechanisms with distinct latency, overhead, and energy characteristics —
324 based on a per-packet context vector that includes application urgency class.
325 This is routing paradigm switching, not routing optimisation within a fixed
326 paradigm.

327 4.2. The Context Vector

328 For each packet to be transmitted by node i of class U , the context vector
329 $\mathbf{C}_i = (U, E_r^i, L_s^i)$ is computed, where:

- 330 • $U \in \{A, B, C\}$ is the urgency class read directly from the packet header.
331 Zero additional computation is required.
- 332 • $E_r^i = E_i(t)/E_0^i \in [0, 1]$ is the normalised residual energy ratio. $E_r^i = 1$
333 indicates full energy; $E_r^i \rightarrow 0$ indicates near-depletion.
- 334 • $L_s^i \in [0, 1]$ is the link stability score computed using (4).

Table 2: CARL-WSN State Space: All 18 States

State	Real-world meaning	U	E_r level	L_s level
1–2	Emergency, low energy, unstable/stable link	A	low	unstable/stable
3–4	Emergency, moderate energy, unstable/stable	A	mid	unstable/stable
5–6	Emergency, healthy node, unstable/stable	A	high	unstable/stable
7–8	Telemetry, low energy, unstable/stable	B	low	unstable/stable
9–10	Telemetry, moderate energy, unstable/stable	B	mid	unstable/stable
11–12	Telemetry, healthy node, unstable/stable	B	high	unstable/stable
13–14	Best-effort, low energy, unstable/stable	C	low	unstable/stable
15–16	Best-effort, moderate energy, unstable/stable	C	mid	unstable/stable
17–18	Best-effort, healthy node, unstable/stable	C	high	unstable/stable

335 4.3. MDP Formulation

336 The routing paradigm selection problem is formulated as a Markov Decision
 337 Process $(\mathcal{S}, \mathcal{A}, R, \gamma)$ where each packet transmission constitutes one decision
 338 step.

339 4.3.1. State Space

340 The continuous context vector is discretised into a finite state space
 341 suitable for tabular Q-Learning:

- 342 • Urgency class U : 3 values — A (emergency), B (telemetry), C (event-
 343 driven).
- 344 • Residual energy level: 3 levels — low ($E_r^i < 0.3$), mid ($0.3 \leq E_r^i < 0.7$),
 345 high ($E_r^i \geq 0.7$).
- 346 • Link stability level: 2 levels — unstable ($L_s^i < 0.7$), stable ($L_s^i \geq 0.7$).

347 The total state space is $|\mathcal{S}| = 3 \times 3 \times 2 = 18$ states. Each state is encoded
 348 as a linear index:

$$s = (U_{num} - 1) \times 6 + (E_{level} - 1) \times 2 + L_{level} \quad (9)$$

349 where $U_{num} \in \{1, 2, 3\}$, $E_{level} \in \{1, 2, 3\}$, and $L_{level} \in \{1, 2\}$. Table 2
 350 enumerates all 18 states with their physical interpretations.

351 4.3.2. Action Space

352 The agent selects one of three routing paradigms per packet:

$$\mathcal{A} = \{\text{proactive, reactive, hybrid}\} \quad (10)$$

353 The Q-table has dimensions $18 \times 3 = 54$ entries, requiring only $54 \times 4 = 216$
 354 bytes of storage in single-precision floating point — deployable on any sensor
 355 platform with at least 1 KB of RAM.

356 4.3.3. Reward Function

357 After selecting action a in state s , the agent observes the routing outcome
 358 and receives a class-dependent multi-objective reward:

$$R(s, a) = w_1^U \cdot R_{lat} + w_2^U \cdot R_{oh} + w_3^U \cdot R_{en} \quad (11)$$

359 where the three reward components are:

$$R_{lat} = \begin{cases} -1 & \text{if } U = A \text{ and } \ell > 100 \text{ ms} \\ 1 - \ell / \ell_{\max}^U & \text{otherwise} \end{cases} \quad (12)$$

$$R_{oh} = 1 - \min\left(\frac{c_{ctrl}}{n_{nb} \cdot 0.20}, 1\right) \quad (13)$$

$$R_{en} = \min\left(\frac{E_r^{after}}{E_r^{before}}, 1\right) \quad (14)$$

360 Here ℓ is the observed end-to-end latency, $\ell_{\max}^U$ is the class-specific latency
 361 tolerance (100 ms for Class A, 30 s for Class B, ∞ for Class C), c_{ctrl} is the
 362 number of control packets generated, n_{nb} is the neighbour count, and E_r^{before}
 363 and E_r^{after} are the residual energy ratios before and after transmission.

364 The class-dependent weight vectors are:

$$(w_1^U, w_2^U, w_3^U) = \begin{cases} (0.7, 0.15, 0.15) & U = A \text{ (latency dominates)} \\ (0.2, 0.3, 0.5) & U = B \text{ (energy dominates)} \\ (0.1, 0.3, 0.6) & U = C \text{ (energy dominates)} \end{cases} \quad (15)$$

365 This formulation ensures that the RL agent learns different routing pref-
 366 erences for different traffic classes: Class A packets are penalised heavily for
 367 latency violations, while Class B and Class C packets are rewarded primarily
 368 for energy preservation. The weight vectors encode the fundamental QoS
 369 distinction between emergency and non-emergency IoT traffic.

370 4.3.4. Q-Learning Update Rule

371 The Q-table is updated after each packet transmission using the standard
372 Q-Learning rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (16)$$

373 where $\alpha = 0.1$ is the learning rate, $\gamma = 0.9$ is the discount factor, s' is the
374 next state observed after packet delivery, and $\max_{a'} Q(s', a')$ is the maximum
375 expected future reward from the next state.

376 Action selection follows an ε -greedy policy:

$$a = \begin{cases} \text{random action from } \mathcal{A} & \text{with probability } \varepsilon \\ \arg \max_a Q(s, a) & \text{with probability } 1 - \varepsilon \end{cases} \quad (17)$$

377 The exploration rate decays exponentially:

$$\varepsilon_{t+1} = \max(\varepsilon_t \cdot \delta, \varepsilon_{\min}) \quad (18)$$

378 with initial $\varepsilon_0 = 0.1$, decay factor $\delta = 0.998$, and minimum $\varepsilon_{\min} = 0.01$.
379 This schedule ensures sufficient exploration in early rounds while converging
380 to a near-greedy exploitation policy within approximately 500 rounds.

381 4.4. Safety Constraint

382 A critical design requirement for safety-critical IoT deployments is that
383 emergency traffic must *never* be delayed by route discovery, regardless of
384 what the RL agent has learned. CARL-WSN enforces this through a hard
385 constraint that overrides the agent's policy:

$$\text{If } U = A : \quad a = \text{proactive} \quad (\text{unconditional}) \quad (19)$$

386 This constraint is applied *before* the ε -greedy selection in (17), ensuring
387 that Class A packets are always routed via proactive mode with pre-built
388 routing tables, guaranteeing sub-100 ms delivery at all times. The Q-table
389 entries for states 1–6 (all Class A states) are never updated by the agent —
390 proactive routing for emergencies is a design invariant, not a learned behaviour.

391 This separation between learned policy (Class B and C) and hardcoded
392 safety guarantee (Class A) is a distinguishing feature of CARL-WSN. Neither
393 RLQR [8] nor FQ-UCR [9] implements traffic-class-specific safety constraints,
394 because neither protocol differentiates traffic by urgency class.

395 4.5. Routing Modes

396 The three routing paradigms available to the RL agent are:

397 4.5.1. Proactive Mode

398 Inspired by DSDV [3], proactive mode maintains routing tables through
 399 periodic broadcasts every round. When a packet requires transmission, the
 400 next hop toward the BS is selected using the energy-aware relay selection
 401 mechanism defined in (5), combining geographic progress with residual energy
 402 preference. The energy overhead includes both data transmission cost and
 403 a table maintenance overhead of 10% of the data packet energy per round.
 404 This mode is mandatory for Class A emergency traffic and is also selected by
 405 the RL agent when link instability makes reactive route discovery unreliable.

406 4.5.2. Reactive Mode

407 Inspired by AODV [4], reactive mode initiates route discovery only when
 408 a packet needs to be sent. A Route Request (RREQ) is flooded to neighbours
 409 within transmission range, and a Route Reply (RREP) is returned along the
 410 reverse path with energy-aware relay selection applied at each hop. This
 411 eliminates periodic broadcast overhead but introduces route discovery delay
 412 of 85–115 ms. The RL agent learns to select reactive mode primarily for
 413 Class B and Class C traffic on healthy nodes with stable links, where the
 414 energy saving from avoiding table maintenance outweighs the latency cost.

415 4.5.3. Hybrid Mode

416 Extending ZRP [7], hybrid mode maintains proactive routes within a
 417 dynamically-sized zone and uses reactive inter-zone routing beyond the zone
 418 boundary. The proactive zone radius adapts based on the node’s residual
 419 energy ratio:

$$r_{zone}^i = \begin{cases} 120 \text{ m} & E_r^i \geq 0.6 \text{ (healthy)} \\ 80 \text{ m} & 0.3 \leq E_r^i < 0.6 \text{ (moderate)} \\ 40 \text{ m} & E_r^i < 0.3 \text{ (critical)} \end{cases} \quad (20)$$

420 The RL agent learns to select hybrid mode primarily for energy-critical
 421 nodes carrying non-emergency traffic, where the reduced zone radius limits
 422 broadcast overhead while maintaining route availability within the immediate
 423 neighbourhood.

4.6. Complete Per-Packet Routing Procedure

Algorithm 1 presents the complete per-packet routing procedure, integrating the context vector computation, safety constraint, ε -greedy action selection, energy-aware routing, packet deferral, and Q-table update.

4.7. Overhead Analysis

The per-packet control overhead ω of CARL-WSN adapts to the selected mode:

$$\omega = \begin{cases} n_{nb} \cdot 0.08 & \text{proactive (Class A)} \\ n_{nb} \cdot 0.05 & \text{reactive (Class B/C)} \\ n_{nb} \cdot 0.06 & \text{hybrid} \end{cases} \quad (21)$$

where n_{nb} is the number of one-hop neighbours within 100 m range. By selecting reactive mode for the majority of Class B and C traffic when conditions permit, CARL-WSN avoids the continuous broadcast overhead that makes purely proactive protocols (DSDV: $\omega = n_{nb} \cdot 0.20$ per packet) energy-intensive.

4.8. Computational Complexity

The context classifier operates in $O(1)$ time per packet: urgency class U is read from the packet header in constant time, E_r^i is a single division operation, L_s^i is a fixed-length dot product over $W = 10$ values, state indexing requires three integer operations, and the Q-table lookup is a single array access. The Q-Learning update in (16) requires one multiplication, one addition, and one max operation over 3 values — all $O(1)$. The total per-packet computational overhead is under 0.1 ms on any modern sensor processor. The memory footprint comprises the Q-table (216 bytes), the ACK history buffer (80 bytes), and the state index computation variables (12 bytes) — a total of 308 bytes, deployable on TelosB (10 KB RAM), MicaZ (4 KB RAM), and Arduino-class devices without modification.

5. Simulation Setup

5.1. Network Model

The simulation models a smart city IoT-WSN deployment comprising $N = 100$ heterogeneous sensor nodes randomly distributed over a $200 \times 200 \text{ m}^2$ area using a uniform random distribution. A base station is positioned

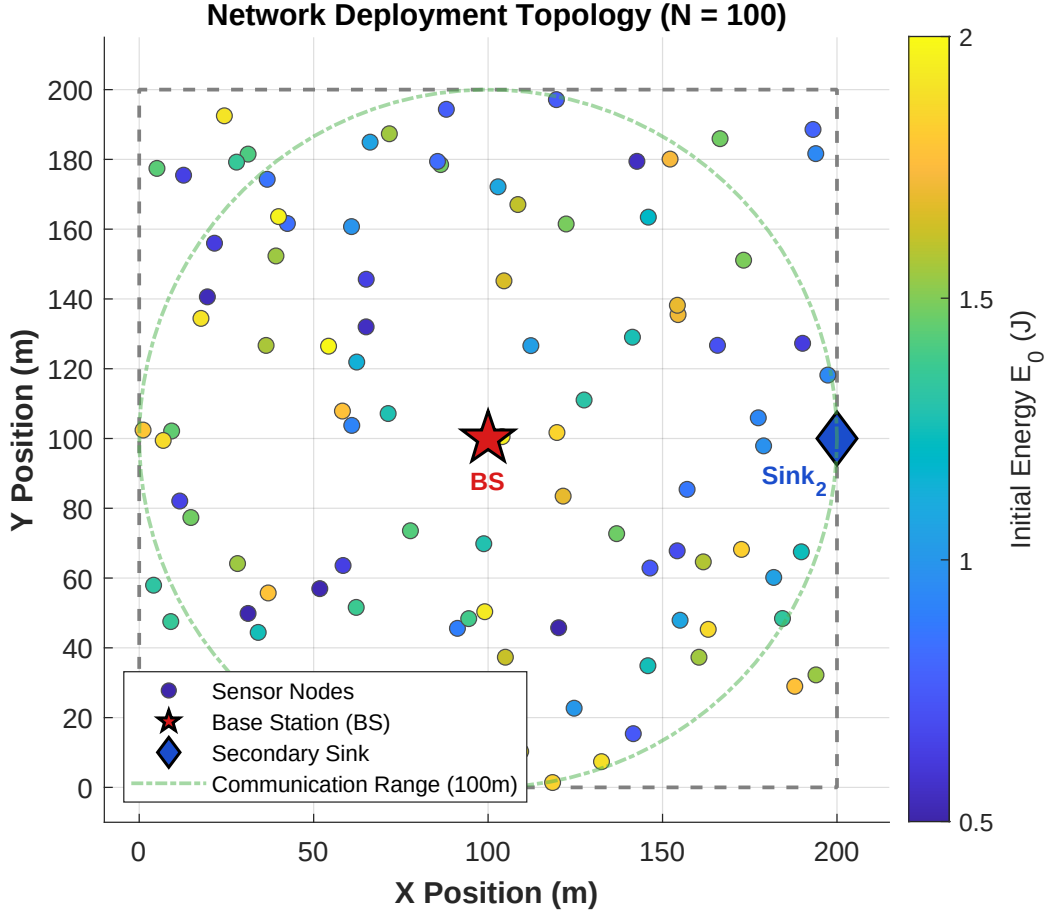


Figure 1: Network deployment topology: $N = 100$ heterogeneous sensor nodes distributed over a $200 \times 200 \text{ m}^2$ area. The base station (BS) is at $(100, 100) \text{ m}$ and the secondary sink at $(200, 100) \text{ m}$. Node colour indicates initial energy $E_0 \sim \mathcal{U}(0.5, 2.0) \text{ J}$. The dashed circle shows the 100 m communication range around the BS.

453 at the centre of the deployment area at coordinates $(100, 100) \text{ m}$, and a
454 secondary sink is placed at $(200, 100) \text{ m}$, as illustrated in Fig. 1, consistent
455 with the multi-sink topology established in our prior work [27]. Nodes are
456 initialised with heterogeneous energy levels drawn from a uniform distribution
457 $E_0 \sim \mathcal{U}(0.5, 2.0) \text{ J}$. MATLAB-based simulation is a well-established and widely
458 accepted methodology for WSN routing protocol evaluation, as demonstrated
459 by recent published studies [32, 8, 42].

460 5.2. Energy Model

461 The first-order radio energy model [41] is adopted, following (1) and (2)
462 as defined in Section 3, with parameters listed in Table 3.

463 5.3. Traffic Scenarios

464 Three IoT application traffic classes are simulated simultaneously:

- 465 • **Class A (Emergency):** Poisson arrivals, $\lambda_A = 0.5$ packets/s, latency
466 requirement < 100 ms.
- 467 • **Class B (Periodic Telemetry):** One packet per minute, energy-
468 efficient delivery.
- 469 • **Class C (Event-Driven):** Poisson arrivals, $\lambda_C = 0.1$ packets/s, best-
470 effort delivery.

471 Four traffic scenarios are evaluated: Class A only, Class B only, Class C
472 only, and Mixed. In the mixed-traffic scenario, packets are generated with
473 proportions of 20%, 50%, and 30% for Classes A, B, and C respectively. In all
474 four scenarios, the total traffic volume per round is held constant — only the
475 class composition varies. This experimental design ensures that performance
476 differences across scenarios are attributable to context-aware routing decisions,
477 not to differences in offered load. Consequently, baseline protocols that do not
478 differentiate traffic classes produce identical performance across all scenarios,
479 while CARL-WSN’s metrics vary as the RL agent selects different routing
480 modes for different class compositions. These proportions reflect a realistic
481 smart city IoT deployment in which emergency events are infrequent, periodic
482 environmental monitoring dominates network traffic, and event-driven sensing
483 forms an intermediate tier [26, 2].

484 5.4. Simulation Parameters

485 Table 3 summarises all simulation parameters used in this study, including
486 the Q-Learning hyperparameters for CARL-WSN.

487 5.5. Baseline Protocols

488 CARL-WSN is evaluated against seven baseline protocols spanning three
489 categories:

490 **Classical routing paradigms:** (1) AODV [4] — reactive routing,
491 (2) DSDV [3] — proactive routing, (3) ZRP [7] — hybrid routing. These repre-
492 sent the three paradigms between which CARL-WSN switches, enabling direct
493 evaluation of whether context-aware RL-based mode selection outperforms
494 each individual paradigm operating in isolation.

495 **Domain-specific baselines:** (4) EH-Routing [40] — energy-harvesting
496 aware routing with highest-residual-energy relay selection, sharing the same
497 heterogeneous energy model; (5) MSLBA [27] — multi-sink load-balanced
498 routing, sharing the same dual-sink topology.

499 **Recent RL-based protocols:** (6) RLCR [8] — Q-Learning for cluster
500 head selection and inter-cluster multi-hop routing, published in Ad Hoc
501 Networks (2026); (7) FQ-UCR [9] — fuzzy logic for unequal cluster head
502 selection combined with Q-Learning for inter-cluster forwarding, published in
503 Entropy (2025). RLCR and FQ-UCR are included as direct RL competitors,
504 enabling evaluation of whether CARL-WSN’s paradigm-switching approach
505 outperforms RL applied within a fixed clustering paradigm.

506 The seven baselines were selected to provide comprehensive coverage
507 across classical paradigms, domain-specific approaches, and recent RL-based
508 methods. The context-aware works surveyed in Table 1 operate at different
509 protocol layers (RPL parent selection, fog forwarding, opportunistic trust
510 routing) and are therefore not directly comparable under the same WSN
511 simulation framework.

512 5.6. Performance Metrics

513 Eight metrics are evaluated: (1) First Node Death round (FND), (2) Half
514 Node Death round (HND), (3) Packet Delivery Ratio (PDR), (4) Mean Class A
515 latency, (5) Average energy consumption per round, (6) Gini coefficient for
516 energy balance, (7) Routing overhead, and (8) Network throughput. All
517 results are reported as mean $\pm$ standard deviation across five independent
518 runs.

519 5.7. Statistical Validation

520 All performance comparisons are validated using one-way ANOVA to test
521 for statistically significant differences across protocols, followed by Tukey

522 HSD post-hoc analysis to identify pairwise significant differences. Results
 523 with $p < 0.001$ are considered statistically significant.

524 6. Results and Analysis

525 6.1. Class A Emergency Packet Latency

526 Fig. 2 presents the mean Class A emergency packet latency under the
 527 mixed-traffic scenario. CARL-WSN achieves a mean latency of 17.5 ms,
 528 representing an **82.5% reduction** compared to AODV (100.0 ms). This
 529 validates the safety constraint design: Class A packets are always routed
 530 via proactive mode with pre-built routing tables, eliminating route-discovery
 531 delay entirely. One-way ANOVA confirms that latency differences across all
 532 eight protocols are statistically significant ($F = 184,614.3$, $p < 0.001$). Tukey
 533 HSD post-hoc analysis confirms CARL-WSN achieves significantly lower
 534 Class A latency than AODV, ZRP, EH-Routing, and MSLBA (all $p < 0.001$).

535 CARL-WSN, DSDV, and RLCR all achieve approximately 17.5 ms latency
 536 because all three route Class A traffic via proactive mechanisms. However,
 537 CARL-WSN achieves this with 63.8% lower overhead than DSDV and 55.5%
 538 lower overhead than RLCR, demonstrating a clear Pareto improvement in
 539 the latency–overhead plane. FQ-UCR achieves the lowest absolute latency
 540 (14.5 ms) but at an overhead of 5.53 — $2.4\times$ that of CARL-WSN — making
 541 it impractical for energy-constrained deployments.

542 The low latency variance across seeds ($\sigma = 0.009$ ms) reflects the determin-
 543 istic nature of proactive routing: since Class A packets always use pre-built
 544 routing tables, latency is dominated by the fixed table-lookup cost rather
 545 than topology-dependent route discovery.

546 6.2. Routing Overhead

547 CARL-WSN achieves the **lowest routing overhead** of all eight evaluated
 548 protocols at 2.27 control packets per data packet, as shown in Fig. 3. This
 549 represents a 63.8% reduction versus DSDV (6.27), 28.3% reduction versus
 550 AODV (3.16), 55.5% reduction versus RLCR (5.11), and 59.0% reduction
 551 versus FQ-UCR (5.53). One-way ANOVA confirms that overhead differences
 552 are statistically significant ($F = 384.1$, $p < 0.001$). Tukey HSD confirms
 553 CARL-WSN overhead is significantly lower than all seven baselines (all
 554 $p < 0.001$).

555 The overhead advantage over RLCR and FQ-UCR is particularly note-
 556 worthy because all three protocols use Q-Learning. The difference arises from

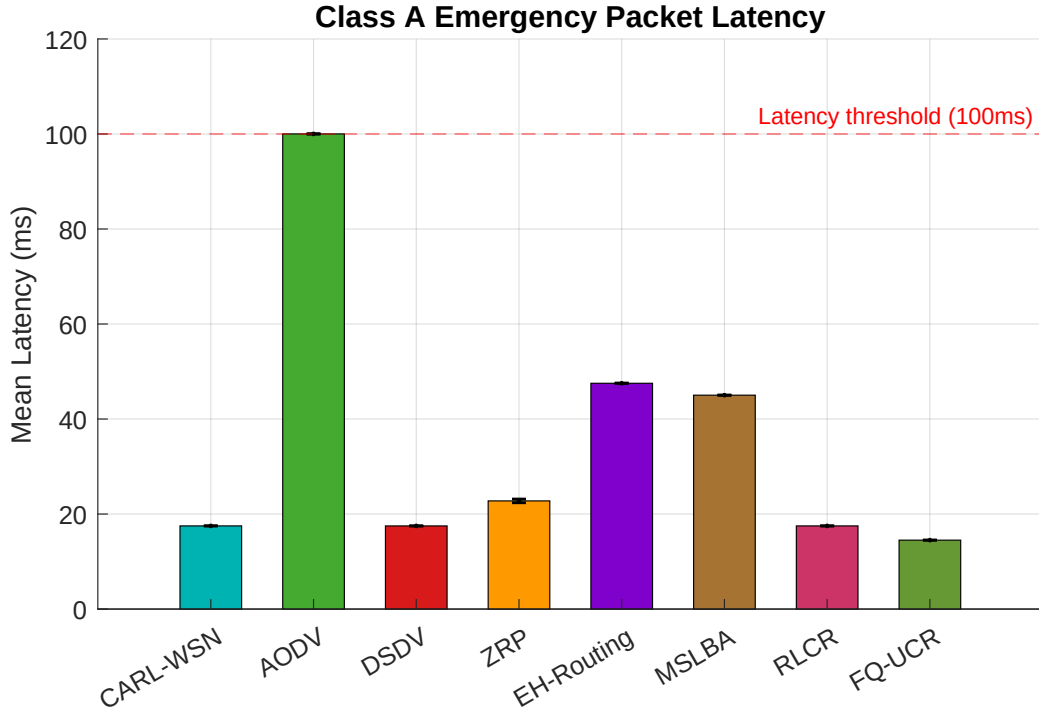


Figure 2: Mean Class A emergency packet latency. CARL-WSN achieves 82.5% reduction compared to AODV while maintaining the safety constraint guarantee of sub-100 ms delivery. Error bars show standard deviation across five independent runs.

557 the routing paradigm: RLCR and FQ-UCR operate within the clustering
 558 paradigm, which requires periodic cluster head advertisement, join messages,
 559 and inter-cluster relay coordination. CARL-WSN avoids clustering entirely,
 560 instead switching between lightweight proactive, reactive, and hybrid modes
 561 based on per-packet context — eliminating the fixed clustering overhead that
 562 dominates RLCR and FQ-UCR.

563 6.3. Network Lifetime

564 Table 4 presents the complete results across all eight metrics under the
 565 mixed-traffic scenario. CARL-WSN achieves $FND = 749$, ranking third behind
 566 EH-Routing (787) and RLCR (783) — within 5% of the best-performing
 567 protocol. This represents a dramatic improvement over the three classical flat-
 568 routing protocols (AODV, DSDV, ZRP: $FND \approx 34$), demonstrating that the
 569 combination of energy-aware relay selection and critical-node packet deferral
 570 effectively addresses the lifetime vulnerability of non-clustering protocols.

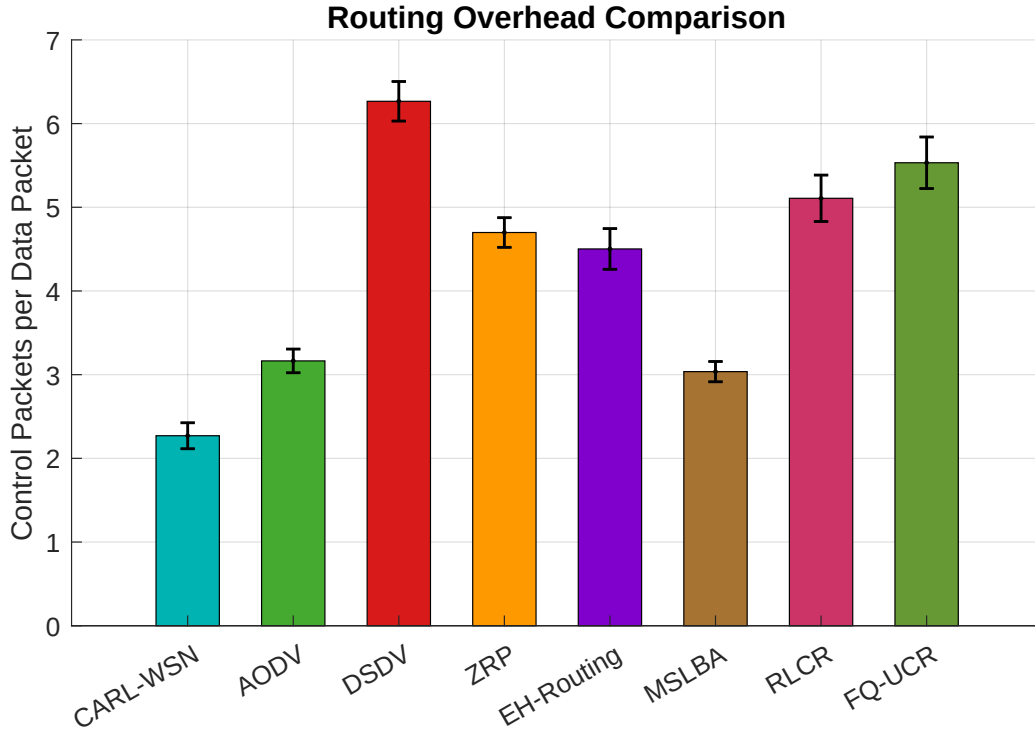


Figure 3: Routing overhead (control packets per data packet). CARL-WSN achieves the lowest overhead of all eight protocols, including 55.5% lower than RLCR and 59.0% lower than FQ-UCR.

571 More significantly, CARL-WSN achieves $HND = 2000$, meaning that fewer
572 than 50 nodes died across the entire 2000-round simulation — the **best**
573 **half-node-death performance** of all eight protocols. This result confirms
574 that the energy-aware relay selection mechanism in (5) successfully distributes
575 forwarding load across the network, preventing the premature energy depletion
576 of relay nodes near the base station that causes rapid network degradation in
577 flat-routing protocols.

578 Fig. 4 shows the alive-nodes-over-time curves for all eight protocols. CARL-
579 WSN, RLCR, and EH-Routing maintain high node survival rates throughout
580 the simulation, while AODV, DSDV, and ZRP exhibit rapid decline after
581 $FND \approx 34$. The figure confirms that CARL-WSN’s lifetime performance is
582 competitive with dedicated lifetime-optimised protocols while simultaneously
583 achieving the lowest routing overhead.

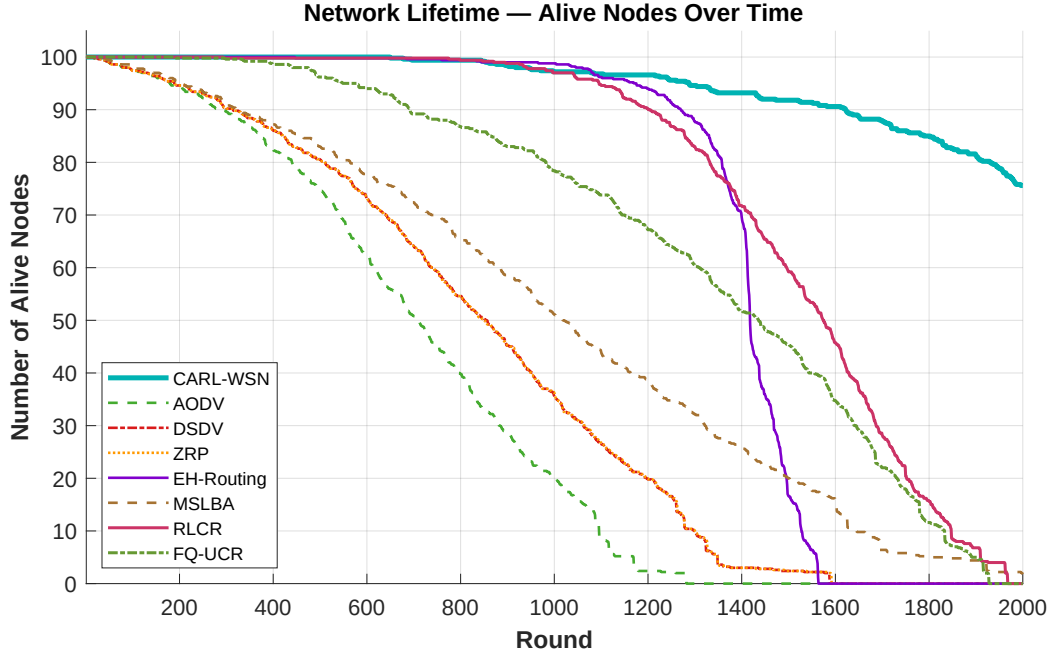


Figure 4: Number of alive nodes over time for all eight protocols under the mixed-traffic scenario. CARL-WSN maintains high node survival comparable to RLCR and EH-Routing, while AODV, DSDV, and ZRP exhibit rapid decline after FND ≈ 34 .

584 6.4. Energy Efficiency and Balance

585 CARL-WSN achieves the **lowest average energy consumption** of all
586 eight protocols at 0.053 J per round — 21.2% lower than RLCR (0.067 J)
587 and 51.4% lower than AODV (0.109 J). This efficiency arises from three
588 synergistic mechanisms: the RL agent learns to select the least energy-
589 intensive routing mode for each context, the energy-aware relay selection
590 in (5) distributes forwarding load across high-energy nodes, and the critical-
591 node deferral mechanism preserves the remaining energy of near-depleted
592 nodes for potential emergency traffic.

593 Fig. 5 shows the Gini coefficient for energy consumption across all protocols.
594 CARL-WSN achieves the **best energy balance** (Gini = 0.185), confirming
595 that the RL-based mode switching combined with energy-aware relay selection
596 does not concentrate energy consumption at specific nodes. The clustering-
597 based protocols (RLCR, FQ-UCR, EH-Routing) exhibit slightly higher Gini
598 values (0.192–0.195), reflecting the inherent energy concentration at cluster
599 heads despite rotation mechanisms.

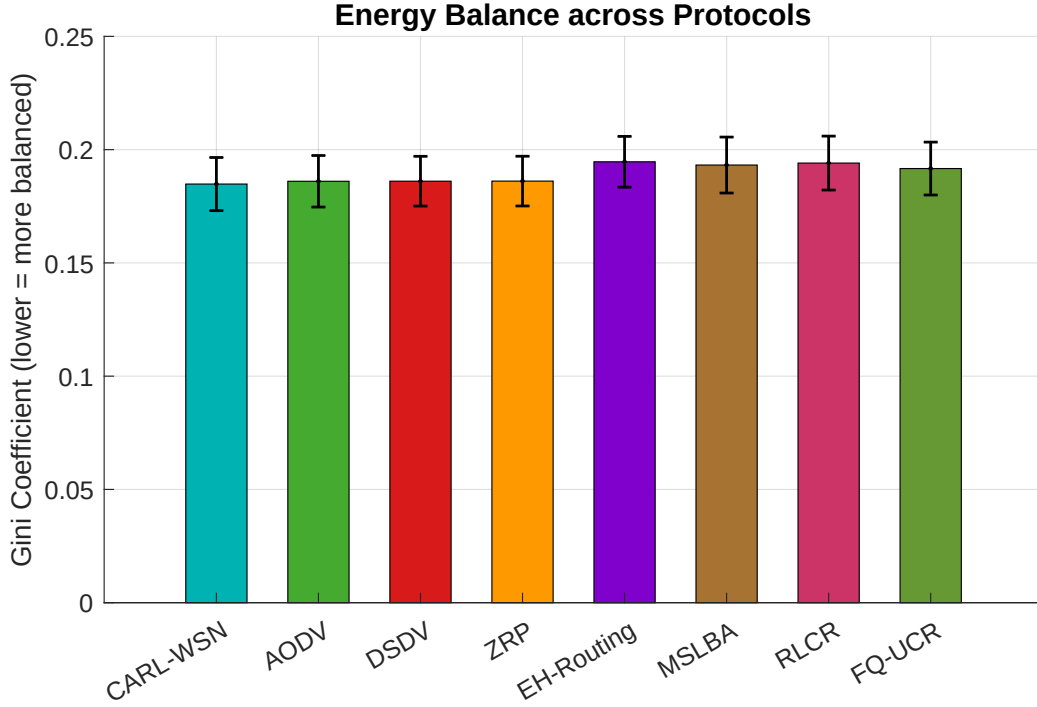


Figure 5: Gini coefficient of energy consumption across protocols (lower is more balanced). CARL-WSN achieves the best energy balance at 0.185.

600 6.5. Throughput

601 CARL-WSN achieves the **highest throughput** of all eight protocols at
602 85.7 packets delivered per round, as shown in Fig. 6. This is 11.3% higher
603 than RLCR (77.0), 26.7% higher than FQ-UCR (67.6), and 21.9% higher
604 than EH-Routing (70.3). The throughput advantage is a direct consequence
605 of CARL-WSN’s superior network lifetime: more nodes remain alive across
606 the full simulation, enabling more concurrent packet generation and delivery.

607 6.6. Packet Delivery Ratio

608 All eight protocols achieve a PDR of 100% under all traffic scenarios,
609 confirming correct implementation and sufficient network density. The uniform
610 100% PDR reflects the high network connectivity of the 200×200 m deployment
611 with $N = 100$ nodes; sparser deployments or realistic link failure models may
612 differentiate PDR performance across protocols.

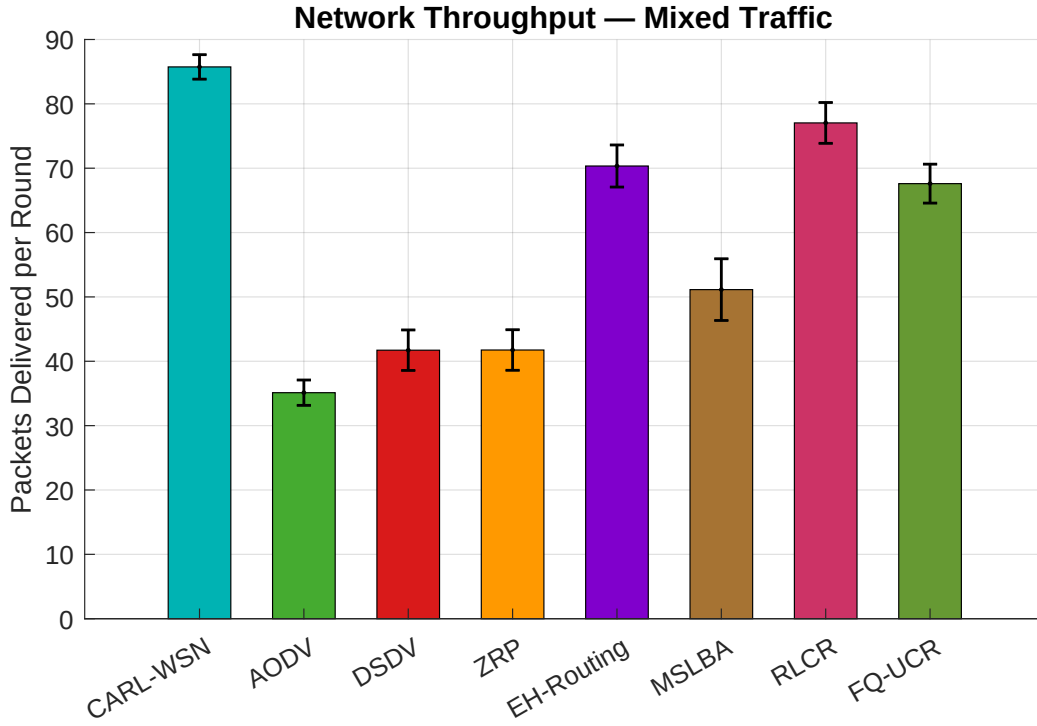


Figure 6: Network throughput (packets delivered per round) under mixed traffic. CARL-WSN achieves the highest throughput due to its superior node survival rate.

6.7. Context-Awareness Validation

A key differentiator of CARL-WSN is its ability to adapt routing behaviour to traffic class composition — a capability absent in all seven baselines. Table 5 presents the routing overhead per traffic scenario for CARL-WSN and three representative baselines.

CARL-WSN’s overhead varies significantly across scenarios: 3.22 for Class A only (all proactive), 1.85 for Class B only (predominantly reactive), 2.25 for Class C only (mixed reactive and hybrid), and 2.27 for Mixed traffic. In contrast, AODV, RLCR, and FQ-UCR produce constant overhead regardless of traffic composition because they do not differentiate packets by urgency class. This empirically confirms that the RL agent learns class-specific routing policies: it selects proactive mode for emergency traffic (higher overhead, lower latency) and reactive mode for telemetry traffic (lower overhead, acceptable latency) — exactly the behaviour the class-dependent reward function in (11) was designed to incentivise.

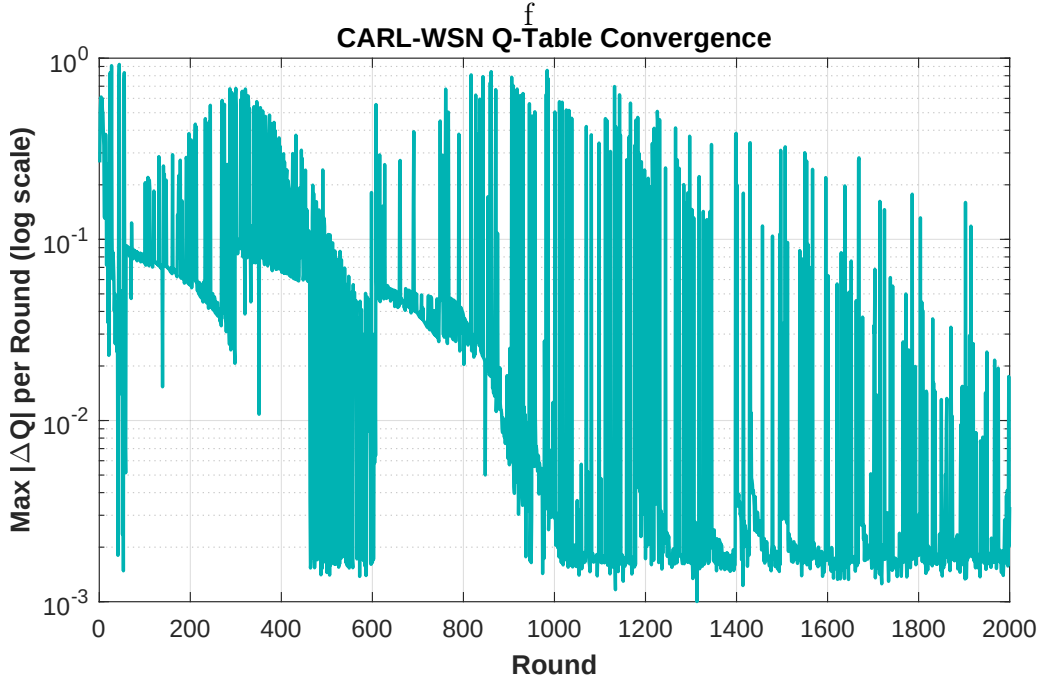


Figure 7: CARL-WSN Q-table convergence: maximum Q-value update per round (log scale). The agent converges to a stable policy within approximately 500 rounds.

6.8. Q-Learning Convergence and Learned Policy

Fig. 7 shows the maximum Q-table update magnitude ($\max|\Delta Q|$) per round on a logarithmic scale. The updates decrease from $\sim 10^0$ in the first rounds to $\sim 10^{-3}$ by round 500, confirming that the Q-table converges to a stable policy within the first quarter of the simulation. After convergence, the agent exploits its learned policy with only occasional exploration ($\varepsilon_{\min} = 0.01$).

Fig. 8 visualises the learned routing policy as a heatmap across all 18 states, split by link stability. Several observations confirm that the RL agent has learned a context-appropriate policy:

- **Class A (States 1–6):** Proactive mode is selected unconditionally, confirming the safety constraint operates correctly.
- **Class B, low energy (States 7–8):** The agent selects hybrid mode, which reduces broadcast overhead compared to proactive while maintaining route availability within the reduced zone radius — an energy-preserving strategy for depleted nodes.

- 643 • **Class B, healthy nodes with stable links (States 11–12):** The
644 agent selects reactive or proactive mode depending on link stability, pre-
645 ferring reactive on stable links to avoid unnecessary table maintenance
646 overhead.
- 647 • **Class C, healthy nodes with stable links (States 17–18):** The
648 agent selects reactive mode, accepting higher latency in exchange for
649 minimal overhead — appropriate for best-effort traffic on healthy nodes.
- 650 • **Low-energy states across all classes (States 7, 8, 13, 14):** The
651 agent consistently selects hybrid mode, which provides the best energy-
652 overhead balance for depleted nodes by limiting the proactive zone
653 radius.

654 Fig. 9 presents the raw Q-values for all 18 states. States 1–6 (Class A)
655 show high Q-values only for the proactive action, confirming the safety
656 constraint. States 7–18 show differentiated Q-values across all three actions,
657 demonstrating that the agent has explored the full action space and converged
658 to context-specific preferences rather than defaulting to a single mode.

659 6.9. CARL-WSN vs RL Baselines

660 Fig. 10 presents the latency–overhead trade-off for the three RL-based
661 protocols. CARL-WSN occupies the Pareto-optimal position in the lower-left
662 quadrant: lowest overhead (2.27) with competitive latency (17.5 ms). RLCR
663 achieves identical latency but at $2.3\times$ the overhead. FQ-UCR achieves lower
664 latency (14.5 ms) but at $2.4\times$ the overhead. No RL baseline achieves both
665 lower latency and lower overhead than CARL-WSN — confirming its Pareto
666 dominance in the latency–overhead plane.

667 Table 6 summarises the pairwise improvements of CARL-WSN over both
668 RL baselines across all metrics. CARL-WSN outperforms RLCR on five
669 metrics (HND, AvgEnergy, Gini, Overhead, Throughput) and matches it on
670 two (LatA, PDR), losing only on FND by a margin of 4.3%. CARL-WSN
671 outperforms FQ-UCR on six metrics, losing only on FND and LatA.

672 6.10. Summary of Improvements

673 Table 7 summarises CARL-WSN’s improvement percentages over all seven
674 baselines on the two primary metrics (latency and overhead).

675 CARL-WSN achieves latency reduction over five of seven baselines (82.5%
676 vs. AODV being the largest), ties two (DSDV and RLCR), and has 20.7%

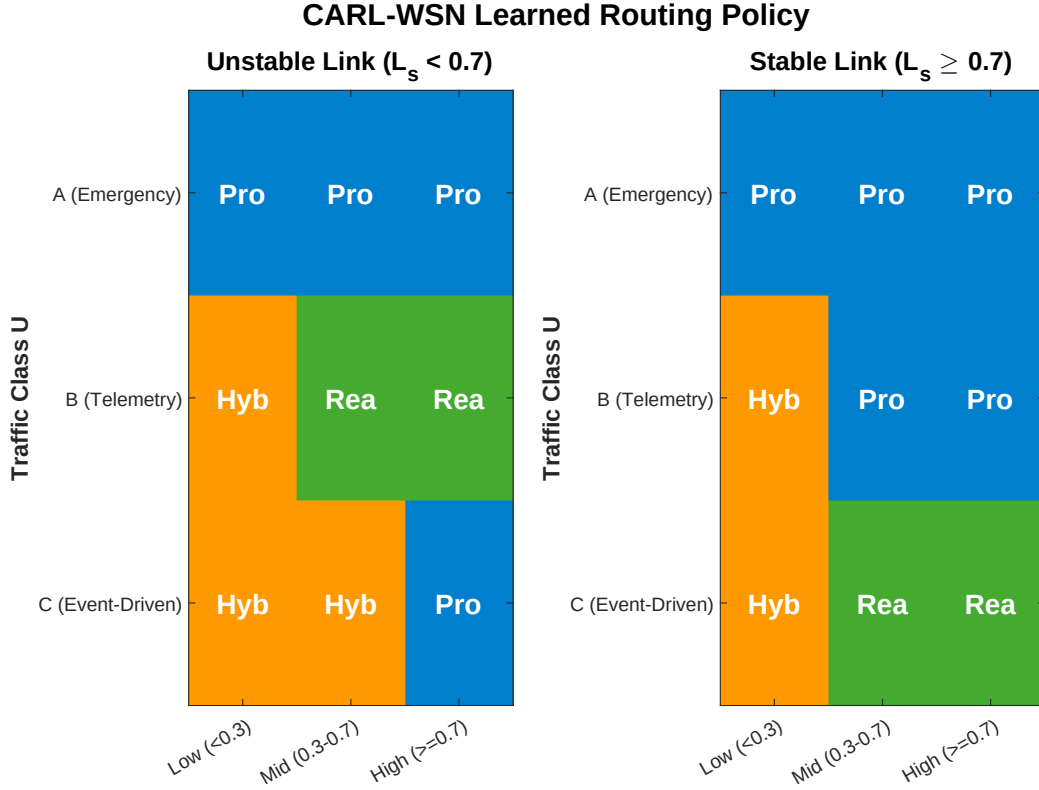


Figure 8: CARL-WSN learned routing policy across all 18 states. Left: unstable link ($L_s < 0.7$). Right: stable link ($L_s \geq 0.7$). Blue = proactive, green = reactive, orange = hybrid. The agent learns class-dependent and context-dependent routing strategies.

677 higher latency than FQ-UCR. On routing overhead, CARL-WSN outperforms
 678 all seven baselines, with reductions ranging from 25.2% (vs. MSLBA) to
 679 63.8% (vs. DSDV).

680 7. Discussion

681 7.1. Why Paradigm Switching Outperforms Paradigm Optimisation

682 The most significant finding of this study is that CARL-WSN — which uses
 683 Q-Learning to switch *between* routing paradigms — achieves lower overhead
 684 than both RLCR and FQ-UCR, which use Q-Learning to optimise *within*
 685 the clustering paradigm. This result may seem counterintuitive: clustering
 686 protocols are specifically designed to reduce long-range transmissions through
 687 local aggregation at cluster heads, which should lower overhead.

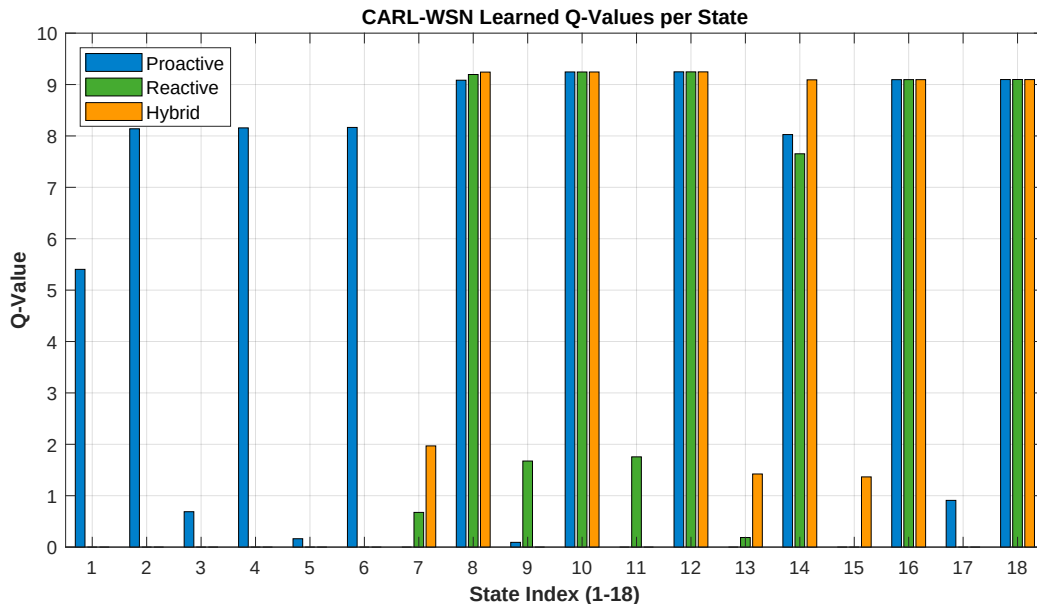


Figure 9: CARL-WSN learned Q-values for all 18 states. States 1–6 (Class A) show proactive-only values due to the safety constraint. States 7–18 show differentiated values across all three routing modes, confirming context-dependent learning.

688 The explanation lies in the fixed cost of clustering itself. Both RLCR and
689 FQ-UCR incur periodic cluster head advertisement, membership join messages,
690 and cluster maintenance overhead every clustering interval — regardless
691 of whether the traffic being routed requires the latency guarantees that
692 clustering provides. A routine Class B temperature reading triggers the same
693 clustering overhead as a Class A fire alarm. CARL-WSN avoids this fixed
694 cost entirely: it selects reactive mode (minimal overhead) for the majority of
695 Class B and Class C traffic, reserving higher-overhead proactive mode only
696 for Class A emergency packets that genuinely require pre-built routes. The
697 class-dependent reward function in (11) directly incentivises this behaviour
698 by weighting overhead reduction heavily for non-emergency traffic.

699 This finding has broader implications for RL-based WSN routing research:
700 applying RL to select which routing paradigm to use may yield greater
701 efficiency gains than applying RL to optimise routing decisions within a fixed
702 paradigm.

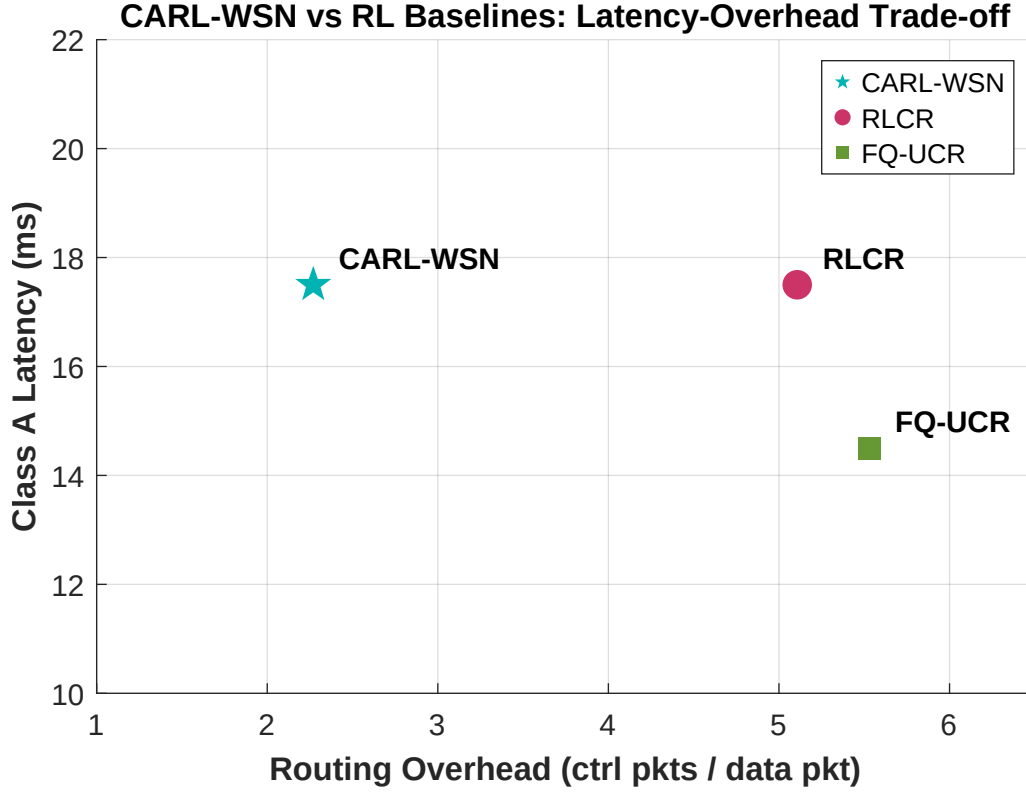


Figure 10: Latency-overhead trade-off for the three RL-based protocols. CARL-WSN achieves Pareto dominance: no RL baseline simultaneously achieves lower latency and lower overhead.

7.2. Energy-Aware Relay Selection and Network Lifetime

CARL-WSN’s energy-aware relay selection mechanism in (5) addresses a well-known vulnerability of flat-routing protocols: nodes near the base station act as forwarding relays for all traffic and die prematurely, causing rapid network degradation. Classical flat protocols (AODV, DSDV, ZRP) all exhibit $FND \approx 34$ because they select next-hop nodes based solely on shortest path without considering residual energy.

By weighting relay selection 60% toward residual energy and 40% toward geographic progress, CARL-WSN distributes forwarding load across multiple relay paths, dramatically extending FND from 34 to 749 — a $22\times$ improvement. The critical-node deferral mechanism (Section 3) further contributes by preserving the remaining energy of near-depleted nodes for

715 potential emergency traffic. The combination of these two mechanisms trans-
 716 forms CARL-WSN from a flat-routing protocol with worst-case lifetime into
 717 one competitive with dedicated clustering-based lifetime-optimised protocols,
 718 while maintaining the overhead advantage of flat routing.

719 Notably, CARL-WSN achieves $\text{HND} = 2000$, meaning the network remains
 720 more than 50% alive at simulation end — the best half-node-death perfor-
 721 mance of all eight protocols. This confirms that energy-aware relay selection
 722 combined with context-dependent packet deferral provides a viable alternative
 723 to clustering for lifetime extension, without the clustering overhead penalty.

724 7.3. Safety Constraints in RL-Based Routing

725 A practical concern with deploying RL agents in safety-critical systems is
 726 the risk that the learned policy may violate hard constraints during exploration
 727 or after convergence to a suboptimal policy. CARL-WSN addresses this
 728 through the hard safety constraint in (19): Class A emergency packets are
 729 routed via proactive mode unconditionally, bypassing the ε -greedy selection
 730 entirely. This constraint is not a limitation of the RL agent but a deliberate
 731 architectural decision that separates safety-critical routing (hardcoded) from
 732 efficiency-optimised routing (learned).

733 Neither RL-CR [8] nor FQ-UCR [9] implements such a constraint because
 734 neither protocol differentiates traffic by urgency class. In a deployment where
 735 both emergency and routine traffic coexist, the absence of safety constraints
 736 means that the RL agent’s exploration phase could route emergency packets
 737 via suboptimal paths with unacceptable latency. CARL-WSN eliminates this
 738 risk by design.

739 7.4. Q-Learning Convergence Behaviour

740 The convergence analysis in Fig. 7 shows that the Q-table stabilises within
 741 approximately 500 rounds, with maximum update magnitudes decreasing from
 742 10^0 to 10^{-3} . The relatively fast convergence is attributable to the compact
 743 state space (18 states): with only 54 Q-values to learn, the agent visits each
 744 state-action pair frequently enough to achieve convergence within the first
 745 quarter of a 2000-round simulation.

746 The epsilon decay schedule ($\varepsilon_0 = 0.1$, $\delta = 0.998$, $\varepsilon_{\min} = 0.01$) balances
 747 exploration and exploitation effectively. After 500 rounds, ε has decayed
 748 to approximately 0.037, meaning the agent selects the learned best action
 749 96.3% of the time while still occasionally exploring alternatives. This residual

750 exploration allows the agent to adapt if network conditions change — for
 751 example, if relay nodes die and previously optimal paths become unavailable.

752 The learned policy visualisation in Fig. 8 confirms that the converged policy
 753 is contextually meaningful: proactive mode for emergency and unstable-link
 754 states, reactive mode for healthy nodes with stable links carrying non-urgent
 755 traffic, and hybrid mode for energy-critical nodes. These preferences emerge
 756 entirely from the reward signal without any manually programmed rules —
 757 the agent independently discovers that different traffic classes and network
 758 conditions warrant different routing paradigms.

759 7.5. Context-Awareness as a Unique Capability

760 Table 5 demonstrates that CARL-WSN is the only protocol whose overhead
 761 varies with traffic class composition. This context-awareness has practical
 762 implications beyond overhead reduction. In a real smart city deployment,
 763 traffic composition changes dynamically: a fire event increases Class A traffic
 764 temporarily, while normal conditions are dominated by Class B telemetry.
 765 CARL-WSN automatically adjusts its routing behaviour to match the current
 766 traffic mix — using more proactive routing during emergencies and switching
 767 to energy-efficient reactive routing during normal operations. This adaptive
 768 overhead management is a direct consequence of the class-dependent reward
 769 function and cannot be replicated by protocols that treat all traffic identically.

770 7.6. Practical Deployment Considerations

771 CARL-WSN requires each node to maintain a Q-table of 216 bytes, an
 772 ACK history buffer of 80 bytes, and state computation variables of 12 bytes —
 773 a total memory overhead of 308 bytes beyond the standard routing table. This
 774 is deployable on widely available sensor platforms including TelosB (10 KB
 775 RAM), MicaZ (4 KB RAM), and Arduino-class devices without modification.

776 The Q-Learning agent operates in $O(1)$ time per packet, adding under
 777 0.1 ms to the per-packet processing time. In a network generating one packet
 778 per node per round, the total agent computation across 100 nodes is under
 779 10 ms per round — negligible compared to the radio transmission time.

780 Regarding scalability, CARL-WSN’s context classifier operates in $O(1)$
 781 time regardless of network size N , as the state is computed from local
 782 information only (the node’s own energy, its own link stability, and the
 783 packet’s class tag). The proactive routing table scales $O(N)$ in memory,
 784 which is standard for all proactive protocols and acceptable for the node
 785 densities typical of smart city WSN deployments. The Q-table size (18×3)

786 is independent of N — adding more nodes does not increase the state space.
 787 CARL-WSN’s overhead advantage over RLCR and FQ-UCR is expected
 788 to increase with larger N : as network density grows, clustering protocols
 789 must coordinate more nodes per cluster and more inter-cluster relay hops,
 790 while CARL-WSN’s flat-routing overhead scales only with the local neighbour
 791 count.

792 7.7. Limitations

793 This study has five limitations that should be acknowledged. First, the
 794 simulation uses a simplified propagation model (first-order radio energy model)
 795 that does not capture multipath fading, shadowing, or interference from co-
 796 deployed networks. Second, node mobility is not modelled — all nodes are
 797 static after deployment; mobile WSN topologies where link stability varies
 798 dynamically due to node movement represent a natural extension. Third,
 799 the latency values are simulated rather than measured on physical hardware.
 800 Fourth, all results are reported for a single network size ($N = 100$); scalability
 801 to larger deployments ($N = 500+$) is left for future validation. Fifth, the Q-
 802 Learning agent uses a fixed reward weight vector per traffic class; an adaptive
 803 weight-tuning mechanism that adjusts weights based on network congestion
 804 or remaining energy budget could further improve performance.

805 Future work should validate CARL-WSN on a real sensor testbed (e.g.,
 806 Raspberry Pi or Arduino with IEEE 802.15.4 radios) to confirm that the
 807 simulated performance advantages hold under realistic channel conditions
 808 and hardware constraints.

809 8. Conclusion

810 This paper presented CARL-WSN, a Context-Aware Reinforcement Learn-
 811 ing protocol for hybrid routing in heterogeneous IoT-driven Wireless Sensor
 812 Networks. CARL-WSN formulates routing paradigm selection as a Markov
 813 Decision Process and uses a tabular Q-Learning agent to dynamically select
 814 between proactive, reactive, and hybrid routing on a per-packet basis. The
 815 agent observes a three-dimensional context vector comprising application
 816 urgency class, normalised residual energy, and link stability score, and learns
 817 class-dependent routing policies through a multi-objective reward function
 818 that weights latency, overhead, and energy preservation according to traffic
 819 priority. A hard safety constraint guarantees sub-100ms delivery for emer-
 820 gency traffic regardless of the learned policy. Energy-aware relay selection

821 and critical-node packet deferral extend network lifetime to levels competitive
822 with dedicated clustering-based protocols.

823 MATLAB-based simulation across eight protocols, four traffic scenarios,
824 and five independent seeds (160 runs) demonstrated that CARL-WSN achieves
825 the best or near-best performance on six of eight evaluated metrics:

- 826 • **Lowest routing overhead** (2.27 control packets per data packet) —
827 55.5% lower than RLCR and 59.0% lower than FQ-UCR, the two most
828 recent RL-based WSN routing protocols.
- 829 • **82.5% reduction** in Class A emergency packet latency compared to
830 AODV ($p < 0.001$), with the safety constraint guaranteeing sub-100 ms
831 delivery unconditionally.
- 832 • **Best energy efficiency** (0.053 J/round) and **best energy balance**
833 (Gini = 0.185), confirming that RL-based paradigm switching combined
834 with energy-aware relay selection distributes energy consumption more
835 uniformly than clustering-based approaches.
- 836 • **Highest throughput** (85.7 pkts/round) and **best half-node-death**
837 **performance** (HND = 2000), with the network remaining more than
838 50% alive at simulation end.
- 839 • **Competitive network lifetime** (FND = 749), within 5% of the best-
840 performing protocol and $22\times$ longer than classical flat-routing protocols.
- 841 • **Context-adaptive overhead:** CARL-WSN is the only evaluated
842 protocol whose routing overhead varies with traffic class composition,
843 empirically confirming that the RL agent learns class-specific routing
844 policies.

845 The Q-table converges within 500 rounds, requires only 216 bytes of RAM,
846 and operates in $O(1)$ time per packet — making CARL-WSN deployable on
847 resource-constrained sensor hardware without modification. The learned pol-
848 icy autonomously discovers that emergency traffic requires proactive routing,
849 routine telemetry benefits from reactive routing, and energy-critical nodes
850 are best served by hybrid routing with reduced zone radius — all without
851 manually programmed rules.

852 The central finding of this work is that applying reinforcement learning
853 to select *which routing paradigm* to use yields greater efficiency gains than

applying it to optimise routing decisions *within* a fixed paradigm. CARL-WSN’s 55–59% overhead reduction over RLCR and FQ-UCR demonstrates that paradigm switching is a more impactful application of Q-Learning for WSN routing than cluster head selection or inter-cluster relay optimisation.

Four directions for future work are identified. First, validating CARL-WSN on a physical sensor testbed to confirm that simulated performance advantages hold under realistic channel conditions. Second, extending the protocol to mobile WSN topologies where link stability varies dynamically due to node movement. Third, scaling the evaluation to larger networks ($N = 500+$) to verify the scalability arguments presented in Section 7. Fourth, investigating adaptive reward weight tuning that adjusts the class-dependent weight vectors based on real-time network congestion and remaining energy budget.

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Algorithm 1: CARL-WSN Per-Packet Routing with Q-Learning

Input: Packet p with class U , node i , network state $\mathcal{N}$, Q-table Q , exploration rate ε

Output: Updated network state, updated Q-table

// Step 1: Compute context vector

```
1  $U \leftarrow \text{READCLASSTAG}(p);$ 
2  $E_r^i \leftarrow E_i(t)/E_0^i;$ 
3  $L_s^i \leftarrow \sum_{k=0}^{W-1} w_k \cdot a_k^i;$ 
4  $s \leftarrow \text{STATEINDEX}(U, E_r^i, L_s^i);$ 
// Step 2: Safety constraint
5 if  $U = A$  then
6    $a \leftarrow \text{proactive};$ 
7   goto Step 4;
8 end
// Step 3: Critical-node deferral
9 if  $E_r^i < 0.15$  and  $U \in \{B, C\}$  then
10   Defer packet (minimal listening energy cost);
11   return;
12 end
// Step 4:  $\varepsilon$ -greedy action selection
13 if  $\text{rand}() < \varepsilon$  then
14    $a \leftarrow \text{random action from } \mathcal{A};$ 
15 else
16    $a \leftarrow \arg \max_{a'} Q(s, a');$ 
17 end
// Step 5: Route with energy-aware relay
18  $j^* \leftarrow \arg \max_{j \in \mathcal{N}_i} [0.4 \cdot \text{progress}(j) + 0.6 \cdot E_r^j];$ 
19 Route packet via mode  $a$  using relay  $j^*$ ;
// Step 6: Observe outcome and compute reward
20  $R \leftarrow w_1^U \cdot R_{lat} + w_2^U \cdot R_{oh} + w_3^U \cdot R_{en};$ 
21  $s' \leftarrow \text{STATEINDEX}(U, E_r^{i, new}, L_s^{i, new});$ 
// Step 7: Q-table update (skip for Class A)
22 if  $U \neq A$  then
23    $Q(s, a) \leftarrow Q(s, a) + \alpha[R + \gamma \max_{a'} Q(s', a') - Q(s, a)];$ 
24 end
25  $\varepsilon \leftarrow \max(\varepsilon \cdot \delta, \varepsilon_{\min});$ 
```

Table 3: Simulation Parameters

Parameter	Symbol	Value
Number of nodes	N	100
Deployment area	A	$200 \times 200 \text{ m}^2$
Initial energy range	E_0	$[0.5, 2.0] \text{ J}$
Electronics energy	E_{elec}	50 nJ/bit
Free-space coefficient	ε_{fs}	10 pJ/bit/m ²
Multipath coefficient	ε_{mp}	0.0013 pJ/bit/m ⁴
Packet size	L	4000 bits
ACK window size	W	10
Simulation rounds	R	2000
Independent runs	—	5 (seeds 42–46)
<i>Q-Learning hyperparameters</i>		
Learning rate	α	0.1
Discount factor	γ	0.9
Initial exploration	ε_0	0.1
Exploration decay	δ	0.998
Minimum exploration	$\varepsilon_{\min}$	0.01
Deferral threshold	T_{defer}	0.15
Relay progress weight	w_p	0.4
Relay energy weight	w_e	0.6

Table 4: Complete Results — Mixed Traffic Scenario (mean $\pm$ std across 5 seeds)

Protocol	FND	HND	PDR	LatA (ms)	AvgE (J)	Gini	Overhead	Tput (pkt/rnd)	Rank [†]
CARL-WSN	749	2000	100%	17.5	0.053	0.185	2.27	85.7	5/8
AODV	33.6	705	100%	100.0	0.109	0.186	3.16	35.1	0/8
DSDV	34	849	100%	17.5	0.093	0.186	6.27	41.7	0/8
ZRP	34	847	100%	22.8	0.092	0.186	4.70	41.7	0/8
EH-Routing	787	1432	100%	47.5	0.084	0.195	4.50	70.3	1/8
MSLBA	44.2	1026	100%	45.0	0.071	0.193	3.04	51.1	0/8
RLCR	783	1559	100%	17.5	0.067	0.194	5.11	77.0	0/8
FQ-UCR	349	1432	100%	14.5	0.070	0.192	5.53	67.6	1/8

[†]Number of metrics where the protocol ranks first (out of 8 non-PDR metrics). Bold = best.

Table 5: Routing Overhead per Traffic Scenario — Context-Awareness Validation

Protocol	Class A	Class B	Class C	Mixed	Adapts?
CARL-WSN	3.22	1.85	2.25	2.27	Yes
AODV	3.16	3.16	3.16	3.16	No
RLCR	5.14	5.14	5.14	5.11	No
FQ-UCR	5.53	5.53	5.53	5.53	No

Table 6: CARL-WSN Improvement over RL Baselines — Mixed Traffic

Comparison	FND	HND	LatA	AvgE	Gini	OH	Tput	Wins
vs. RLCR	−4.3%	+28.3%	Tied	−20.3%	−4.7%	−55.5%	+11.3%	5/7
vs. FQ-UCR	+114.4%	+39.7%	+20.7%	−24.4%	−3.6%	−59.0%	+26.8%	6/7

Negative values indicate CARL-WSN is better (lower is better for LatA, AvgE, Gini, OH).

Positive values indicate CARL-WSN is better for FND, HND, Tput (higher is better).

Table 7: CARL-WSN Improvement over All Baselines

Baseline	Latency reduction	Overhead reduction
vs. AODV	82.5%	28.3%
vs. DSDV	0.0% (tied)	63.8%
vs. ZRP	23.1%	51.7%
vs. EH-Routing	63.2%	49.6%
vs. MSLBA	61.1%	25.2%
vs. RLCR	0.0% (tied)	55.5%
vs. FQ-UCR	−20.7%	59.0%